

DHCP — Guided Notes

Unit 1.6.1 | CompTIA Network+ N10-009 Obj. 1.6 | N10-008 1.6

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A DHCP scope defines the _____ of addresses available for assignment, the _____ time, and options such as _____, _____, and _____. Exclusions are addresses within the scope range that the server will _____.

Question 2

A DHCP reservation assigns a _____ IP address to a specific device based on its _____. This differs from a static IP because the address is still managed by _____ rather than configured manually on the device.

Question 3

When a DHCP lease is _____ % expired, the client attempts to renew with the same server. If the server is unavailable when the lease fully expires, the client falls back to _____ and starts the _____ process over.

Question 4

A DHCP relay agent (helper address) is needed when _____. The relay agent is configured on the _____ and forwards DHCP broadcasts as _____ packets to the DHCP server's IP address.

Question 5

DHCP failover has two modes. In _____ mode, one server handles all leases and the other takes over only if the primary fails. In _____ mode, both servers share the address pool and handle requests simultaneously.

Question 6

Match each DHCP problem to its symptom:

Exhausted scope → _____

Rogue DHCP server → _____

Misconfigured relay → _____

IP address conflict → _____

Question 7

To check available addresses in a Windows DHCP scope, you would use _____. To force a client to release and request a new lease, the commands are _____ followed by _____.

Question 8 — Real World Application

REAL WORLD: On Monday morning, 30 students in one classroom can't get on the network. They all have 169.254.x.x addresses. Students in other classrooms are fine. Walk through your diagnosis in order.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1-6-1-dhcp/1-6-1-dhcp-answer-key.pdf
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